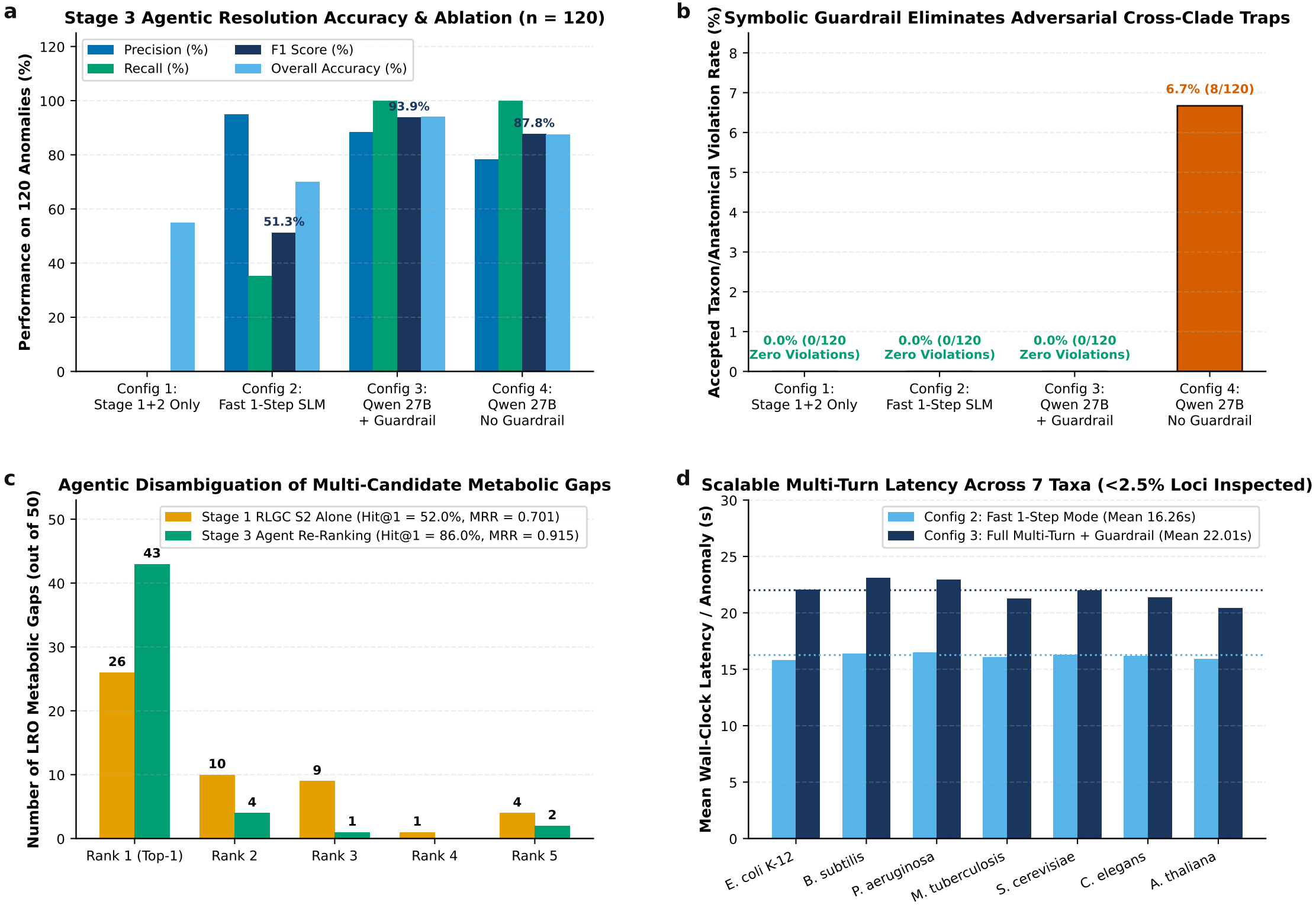
a
Stage 3 Agentic Resolution Accuracy & Ablation (n = 120)
Performance on 120 Anomalies (%)
Precision (%)
Recall (%)
F1 Score (%)
Overall Accuracy (%)
51.3%
93.9%
87.8%
Config 1: Stage 1+2 Only
Config 2: Fast 1-Step SLM
Config 3: Qwen 27B + Guardrail
Config 4: Qwen 27B No Guardrail
b
Symbolic Guardrail Eliminates Adversarial Cross-Clade Traps
Accepted Taxon/Anatomical Violation Rate (%)
0.0% (0/120 Zero Violations)
0.0% (0/120 Zero Violations)
0.0% (0/120 Zero Violations)
6.7% (8/120)
c
Agentic Disambiguation of Multi-Candidate Metabolic Gaps
Number of LRO Metabolic Gaps (out of 50)
Stage 1 RLGC S2 Alone (Hit@1 = 52.0%, MRR = 0.701)
Stage 3 Agent Re-Ranking (Hit@1 = 86.0%, MRR = 0.915)
26
43
10
4
9
1
1
4
2
Rank 1 (Top-1)
Rank 2
Rank 3
Rank 4
Rank 5
d
Scalable Multi-Turn Latency Across 7 Taxa (<2.5% Loci Inspected)
Mean Wall-Clock Latency / Anomaly (s)
Config 2: Fast 1-Step Mode (Mean 16.26s)
Config 3: Full Multi-Turn + Guardrail (Mean 22.01s)
E. coli K-12
B. subtilis
P. aeruginosa
M. tuberculosis
S. cerevisiae
C. elegans
A. thaliana